Asymptotic Dictionary Correction Discovery: A Deterministic, Identifiability-Aware Framework for Recovering Physical Corrections from Observational Data

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Abstract

We present **ADCD** (Asymptotic Dictionary Correction Discovery), a deterministic framework for recovering structured corrections to known physical laws from noisy data. Given a classical baseline, ADCD searches for the residual correction that vanishes exactly at the classical limit, using three interlocking mechanisms: algebraically regularized primitives that enforce asymptotic vanishing as an exact algebraic identity; a five-dimensional (MLTΘQ) Buckingham-π engine that auto-derives dimensionless ratio arguments without manual specification; and a Phenomenon-Specific Taxonomy encoding peer-reviewed functional priors for each physics domain. A four-step validation protocol—budget disclosure, positive control, BIC-based ablation control, and determinism check—issues one of two verdicts: IDENTIFIABLE or WITHHELD. On three canonical scenarios at restricted low-signal windows, Screened Coulomb ($\Delta\mathrm{BIC} = 30.65$, exact match) and Entropy Expansion ($\Delta\mathrm{BIC} = 14.70$, class-level, parsimony-driven; 2/5 seeds) are IDENTIFIABLE; Time Dilation at $v \leq 0.3c$ is WITHHELD because its observational noise floor ($\mathrm{NMSE} \approx 0.41$) exceeds the positive-control threshold—the intended, epistemically honest behavior. All results are byte-exact deterministic.

Keywords: symbolic regression; correction discovery; dimensional analysis; Bayesian prior; model selection; identifiability

1 Introduction

In the expanding landscape of machine learning for the physical sciences Carleo et al. [2019], Wang et al. [2023], physical laws frequently take a correction-first form: a known classical baseline extended by a term that vanishes at the classical limit Einstein [1905], Debye and Hückel [1923], Callen [1985]. Recovering that correction from observational data is distinct from full equation discovery: the baseline is fixed, only the residual must be identified, and that residual must vanish exactly where classical physics holds.

General-purpose symbolic regression (SR) engines such as PySR Cranmer et al. [2020], Cranmer [2023], AI Feynman Udrescu and Tegmark [2020], and PhySO Tenachi et al. [2023] excel at equation discovery from scratch but face three difficulties in the correction-first setting: (i) *catastrophic cancellation*: floating-point precision collapses as $u \to 0$, corrupting optimizer feedback IEEE Task P754 [2019]; (ii) *stochastic irreproducibility*: default evolutionary search does not guarantee byte-identical results across runs without serial execution Bongard and Lipson [2007], Schmidt and Lipson [2009]; and (iii) *silent overclaiming*: a single best-fit equation is returned regardless of whether the data carry sufficient power to distinguish competing structures La Cava et al. [2021], generating

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what recent meta-science critiques identify as an illusion of understanding Messeri and Crockett [2024], Krenn et al. [2022]. ADCD addresses each failure mode in turn: algebraically regularized primitives eliminate cancellation; deterministic grammar enumeration ensures reproducibility; explicit BIC-based identifiability gating prevents overclaiming.

2 Related Work

Automated scientific discovery traces back to pioneering heuristic systems such as BACON Langley et al. [1987] and has evolved toward modern autonomous equation discovery Kramer et al. [2026], Makke and Chawla [2024], Karagiorgi et al. [2022]. General SR engines (Eureqa Schmidt and Lipson [2009], PySR Cranmer et al. [2020], Cranmer [2023], AI Feynman Udrescu and Tegmark [2020], Udrescu et al. [2020], PhySO Tenachi et al. [2023]) target unconstrained equation recovery and do not enforce classical-limit constraints algebraically. SINDy Brunton et al. [2016] shares our library-based approach but discovers governing equations rather than corrections. PINNs Raissi et al. [2019], Karniadakis et al. [2021] embed physical laws as soft loss penalties; ADCD imposes them as hard algebraic identities before any numerical computation. Recent systems combining data with background knowledge and axiomatic reasoning, such as AI-Descartes Cornelio et al. [2023] and AI Hilbert Cory-Wright et al. [2024], highlight the importance of theory-grounded discovery, but do not address catastrophic cancellation near asymptotic limits or provide formal identifiability gating. While dimensional analysis and symmetry constraints have been integrated into SR and dynamical learning Tenachi et al. [2023], Petersen et al. [2021], Otto et al. [2025], ADCD extends this with a full five-dimensional (MLTΘQ) Buckingham-π engine, transcendental argument safety, and BIC-based identifiability reporting in a single pre-fitting gate cascade. To our knowledge, ADCD is the first framework to combine deterministic grammar enumeration, algebraic classical-limit enforcement, phenomenon-specific Bayesian priors, and explicit identifiability gating for the correction-first paradigm.

3 Methods

3.1 Problem formulation

Let y_{obs} and y_{cl} be the observed target and classical prediction. The correction residual is:

$$\Delta = \begin{cases} y_{\text{obs}}/y_{\text{cl}} - 1 & \text{(multiplicative)} \\ y_{\text{obs}} - y_{\text{cl}} & \text{(additive)} \end{cases} \tag{1}$$

The mode with the lower scale dependence is selected, measured as the absolute Spearman correlation between $|y_{\text{cl}}|$ and the residual magnitude (ties go to additive). There is no numeric ambiguity threshold: additive is used by construction when $\text{std}(y_{\text{cl}}) < 10^{-15}$, when fewer than 5 valid points remain, or when both correlations vanish. ADCD searches for $\hat{\Delta}(\mathbf{x}; \boldsymbol{\theta})$ such that $\hat{\Delta} \to 0$ as the relevant dimensionless ratio $u \to 0$.

3.2 Algebraically regularized primitive dictionary

Each primitive $D(u)$ satisfies $\lim_{u\to 0} D(u) = 0$ as an exact algebraic identity by construction, while rationalizing subtraction-prone forms against catastrophic cancellation. For example, the naive form $(1-u)^{-1/2} - 1$ collapses to 0.0 at $u = 10^{-16}$ in `float64`; the rationalized form $u/[\sqrt{1-u}(\sqrt{1-u}+1)]$ returns 5.0×10^{-17} at the same input. Table 1 lists all eight primitives.

3.3 Dimensional grouping and free-scale generation

ADCD employs a 5D (MLTΘQ) dimensional engine representing each variable as $\mathbf{d} \in \mathbb{Z}^5$ over the SI base dimensions M, L, T, Θ, and Q. Rather than computing the full rational nullspace, dimensionless arguments are generated via a two-pass domain heuristic: first, it groups physical variables with identical dimensions into pairwise ratios (e.g., q_1/q_2); second, each dimensional variable is normalized by linear and quadratic free-scale parameters ($v/\theta_{2k}, v^2/\theta_{2k+1}$; e.g., r/θ_4). Extending from the conventional 3D MLT to 5D is essential for correctness: in a 3D engine charges q_1, q_2 are dimensionless scalars and incorrectly bypass the ratio generation; in the 5D engine, charge

Table 1: The eight algebraically regularized primitives, each satisfying $\lim_{u\to 0} D(u) = 0$ as an exact algebraic identity. The first five are the core physics-motivated basis; the remaining three extend coverage to oscillatory, saturation, and power-law regimes. Column "Physical origin" cites the fundamental equation that produces this functional family; it does not imply the primitive is restricted to that domain.

Name	Regularized form $D(u)$	Domain	Physical origin
`D_lor`	$u/[\sqrt{1-u}(\sqrt{1-u}+1)]$	$u \in [0, 1)$	Lorentz factor $(\gamma - 1)$
`D_rat`	$u/(1-u)$	$u \neq 1$	Yukawa/Fermi pole
`D_exp`	$e^{-u} - 1$	all u	Debye/Yukawa screening
`D_log`	$\ln(1+u)$	$u > -1$	Entropy of mixing
`D_sqrt_inv`	$1/\sqrt{1+u} - 1$	$u > -1$	Gravitational / Coulomb potential
`D_pow`	$\lvert u\rvert^{\lvert\theta\rvert}$	$u \geq 0$	Power-law corrections
`D_osc`	$1 - \cos u$	all u	Wave / AC oscillatory
`D_sat`	$\tanh u$	all u	Magnetic / sigmoid saturation

carries genuine Charge dimension (Q), ensuring q_1, q_2 form pairwise ratios in Pass 1 and consume four free-scale parameters (θ_0–θ_3) in Pass 2 before distances are indexed. Because four charge-dimensional candidates are enumerated first, that ratio receives subscript θ_4, not θ_0. This subscript is machine-verifiable evidence of automated discovery: a manually supplied ratio would be pinned to θ_0 by convention, not assigned by enumeration.

3.4 Phenomenon-specific taxonomy as Bayesian prior

Each physics phenomenon maps to a named set of permitted primitive families derived exclusively from its fundamental equations. For example, `yukawa_debye_screening` permits only `D_exp` and `D_rat`: the Klein-Gordon equation yields $V \propto e^{-mr}/r$ and Debye-Hückel theory yields $\phi \propto e^{-r/\lambda_D}/r$; both involve only exponential and rational families. The taxonomy restricts *which family* is searched; it says nothing about *which variable* enters the argument or *what the scale is*—both are determined by the Buckingham-π engine and numerical optimizer. Including a physically unjustified primitive acts as a noise sponge, collapsing the Ablation Control BIC gap and destroying identifiability. Table 2 lists entries for our three experiments.

A 3×3 cross-misspecification audit (Appendix A.3, Table 9) confirms the taxonomy is *safe against misspecification*: in all 6 wrong-taxonomy trials the system returned WITHHELD (0/6 false discoveries; 4 of the 6 are informative, since the 2 Time Dilation trials are WITHHELD even under the correct prior; e.g., applying the oscillatory `wave_resonance` prior to Screened Coulomb yields WITHHELD with $\Delta\text{BIC} = -910.5$).[1] An incorrect prior causes conservative abstention, not hallucinated laws.

Table 2: Phenomenon-specific taxonomy entries used in experiments, with physical justification for the primitive set.

Phenomenon key	Primitives	Justification
`yukawa_debye_screening`	`D_exp`, `D_rat`	Klein-Gordon / Debye-Hückel: fundamental form is e^{-mr}/r.
`lorentz_special_relativity`	`D_lor`	Lorentz invariance of Maxwell equations uniquely selects the $\gamma - 1$ form.
`boltzmann_thermodynamics`	`D_exp`, `D_log`	Boltzmann partition function and entropy of mixing involve $e^{-\beta\epsilon}$ and ln.

3.5 Grammar enumeration and theta-intact constraint

Candidates take the form $\hat{\Delta} = \theta_0 \cdot D_i(u_{ij})$, optionally extended with a secondary additive or multiplicative-coupled term ($\theta_0 D_a(u) + \theta_1 D_b(u)$ or $\theta_0 D_a(u)[1 + \theta_1 D_b(u)]$; depth ≤ 3, ≤ 25 tokens). Dimensionless arguments are auto-generated by the Buckingham-π engine (up to 12

[1] Full supplementary audit tables are provided in Appendix A and at https://github.com/apiprdt/PhysicsPaper/tree/PythonADCD/rebuttal.

candidate ratios per scenario). Gates 1–3 (Table 3) inspect the symbolic *theta-intact* expression, treating single-variable free-scale ratios (e.g., r/θ_4 representing a Debye length) as dimensionally admissible because θ_k absorbs the variable's physical scale during fitting. Gate 4 evaluates the symbolic classical limit at $\boldsymbol{\theta} = \mathbf{1}$ (deferring parametric candidates for strict post-fit re-verification at $\hat{\boldsymbol{\theta}}$), and Gate 5 screens coarse numerical fitness at $\boldsymbol{\theta} = \mathbf{1}$.

3.6 Physical gate cascade

Before fitting, each candidate must pass a five-gate cascade (Table 3) applied to the theta-intact expression.

Table 3: The five-gate pre-fitting physical cascade.

Gate	Rejection criterion
1. AST complexity	SymPy tree depth > 12 or token count > 50.
2. Dimensional consistency	SI base dimensions mismatch target (allowing free-scale v/θ_k).
3. Transcendental safety	Arguments to $\exp$, $\ln$, $\sqrt{\cdot}$ not dimensionless or θ_k-scaled.
4. Asymptotic check (ARC)	Classical limit at $\boldsymbol{\theta} = \mathbf{1}$ (and post-fit $\hat{\boldsymbol{\theta}}$) does not vanish.
5. Coarse numerical fitness	Evaluation at $\boldsymbol{\theta} = \mathbf{1}$ produces `NaN/Inf`.

3.7 Parameter optimization and model selection

The top-ranked Stage-1 surviving candidates ($K = 30$) are optimized via L-BFGS-B (JAX, `float64`). Parameters are log-reparameterized ($\theta_i{=}s_i e^{u_i}$, $s_i{\in}\{-1, +1\}$) for order-of-magnitude traversal; multiple restarts reduce sensitivity to local optima. Model selection uses an Extended BIC Schwarz [1978], Chen and Chen [2008]: $\mathrm{BIC} = k \ln N - 2 \ln \hat{L} + 2 \ln |\mathcal{C}|$, incorporating a multiple-testing penalty over the search space size $|\mathcal{C}|$. By the Kass–Raftery scale Kass and Raftery [1995], $\Delta\mathrm{BIC} > 10$ is the adopted identifiability threshold.

3.8 Four-step validation protocol

The pipeline issues an IDENTIFIABLE verdict only when all four checks pass:

1. **Budget disclosure.** The complete size of the domain-guided search space $|\mathcal{C}|$ and the primitive list are reported before any results are seen.
2. **Positive control.** Restricted to the ground-truth primitive family alone, the system must recover the correct structure (NMSE < 0.05; i.e., less than 5% unexplained variance).
3. **Ablation control.** With the ground-truth family excluded from an otherwise unrestricted search over the full eight-primitive registry, the BIC gap $\Delta\mathrm{BIC} > 10$ (Kass–Raftery "very strong evidence") must hold between the domain-guided Rank-1 candidate from the blind search and the best Rank-1 candidate found under ablation.
4. **Determinism.** Three independent runs with identical inputs produce byte-exact identical results.

Failure of any check yields WITHHELD.

4 Experiments

4.1 Setup

Table 4 summarizes the three scenarios. All use 1% Gaussian multiplicative noise, $N = 200$ points, seed 42. Observation windows reflect historical low-signal regimes. The ground truth is never supplied to the blind discovery search (only noisy observations, classical predictions, dimensional definitions, and the taxonomy key are provided; the target family is referenced only in benchmark controls).

Table 4: Experimental scenarios with domains, historical observation windows, taxonomy keys, targets, and the full domain-guided search space size ($|\mathcal{C}|$).

Scenario	Domain	Window	Taxonomy key	Target	$\lvert\mathcal{C}\rvert$
Time Dilation	Relativity	$v \leq 0.3c$	`lorentz_special_relativity`	`D_lor`	10
Screened Coulomb	Electrostatics	$r \leq 4.0$ m	`yukawa_debye_screening`	`D_exp`	60
Entropy Expansion	Thermodynamics	$dV/V_i \leq 1.0$	`boltzmann_thermodynamics`	`D_log`	40

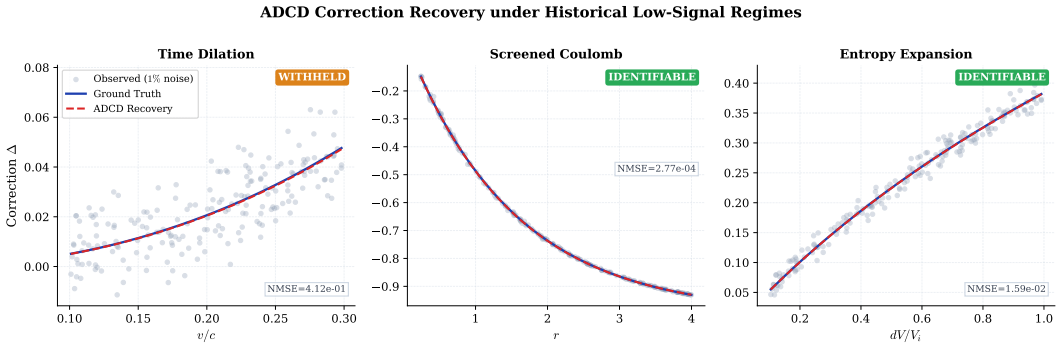


Figure 1: Correction recovery for the three scenarios. *Gray dots*: observed residual Δ at 1% Gaussian noise. *Blue solid*: ground truth. *Red dashed* ("ADCD Recovery"): the exact symbolic expression discovered by ADCD, evaluated with the fitted parameters $\boldsymbol{\theta}$; see Table 5 for exact NMSE values. Screened Coulomb (IDENTIFIABLE): exact symbolic match, $\mathrm{NMSE} = 2.77 \times 10^{-4}$. Entropy Expansion (IDENTIFIABLE): class-level match, $\mathrm{NMSE} = 1.59 \times 10^{-2}$. Time Dilation (WITHHELD): Lorentz structure found at Rank 1, $\mathrm{NMSE} = 0.41$; positive control fails within the historical $v \leq 0.3c$ observation window.

4.2 Results

Table 5 summarizes the quantitative results. Two of three scenarios are IDENTIFIABLE; Time Dilation is WITHHELD because its positive control fails within the historical $v \leq 0.3c$ observation window. ΔD, the change in deviance $D = -2 \ln \hat{L} = N \ln \mathrm{NMSE}$ (twice the unpenalized NLL difference), isolates the pure fit-quality advantage from the multiple-testing penalty, addressing a known asymmetry in Extended BIC when $|\mathcal{C}_{\mathrm{domain}}| \neq |\mathcal{C}_{\mathrm{ablated}}|$.[2]

Table 5: Four-step validation results for all three scenarios. PC = Positive Control (pass/fail). $\Delta\mathrm{BIC} = \mathrm{BIC}_{\mathrm{ablated}} - \mathrm{BIC}_{\mathrm{best}}$, where $\mathrm{BIC}_{\mathrm{best}}$ is the Rank-1 domain-guided search BIC; positive values indicate the full-primitive set is preferred over the ablated set. ΔD: penalty-free fit-quality difference (see footnote). Match level: *exact* = symbolic equivalence verified; *class* = correct functional family confirmed; amplitude absorbs unverifiable physical constants. Seed stability over 5 independent noise realizations: Screened Coulomb 5/5 IDENTIFIABLE; Entropy Expansion 2/5 IDENTIFIABLE (borderline at the Kass–Raftery boundary); Time Dilation 5/5 WITHHELD.

Scenario	Discovered structure	Rank	NMSE	PC	ΔD	ΔBIC	Match	Verdict
Screened Coulomb ($r \leq 4$ m)	$\theta_0(e^{-r/\theta_4} - 1)$	1	2.77×10^{-4}	PASS	20.14	30.65	exact	**IDENTIFIABLE**
Entropy Expansion ($dV/V_i \leq 1$)	$\theta_0 \ln(1 + dV/V_i)$	1	1.59×10^{-2}	PASS	-1.08	14.70	class	**IDENTIFIABLE**
Time Dilation ($v \leq 0.3c$)	$\theta_0 \frac{v^2/c^2}{\sqrt{1-v^2/c^2}(1+\sqrt{1-v^2/c^2})}$	1	0.41	FAIL	-0.00	13.79	class	**WITHHELD**

Screened Coulomb. Recovered $\theta_0(e^{-r/\theta_4} - 1)$ is an exact symbolic match to the ground truth $\theta_0(e^{-r/\lambda_D} - 1)$, with $\theta_4 \approx 1.49$ m recovering the Debye length $\lambda_D = 1.50$ m. Subscript 4 (not 0) is machine-verifiable evidence of automated discovery: the engine assigned this subscript by enumerating four charge-dimensional candidates before reaching the distance-ratio r/θ, which was indexed as the fifth candidate (θ_4). A manually supplied ratio would be pinned to subscript 0 by

[2] ΔBIC includes a search-space penalty $2 \ln |\mathcal{C}|$ that differs between domain-guided and ablated runs; ΔD reports the penalty-free fit-quality difference. Here $|\mathcal{C}_{\mathrm{ablated}}| = 814/534/700$ for Screened Coulomb, Entropy Expansion, and Time Dilation, versus $|\mathcal{C}_{\mathrm{domain}}| = 60/40/10$. Screened Coulomb's $\Delta D = 20.14$ alone exceeds the Kass–Raftery threshold of 10. Entropy Expansion's identifiability is parsimony-driven ($\Delta D = -1.08$; the ablated model achieves an equivalent fit with more parameters), a distinction we report transparently.

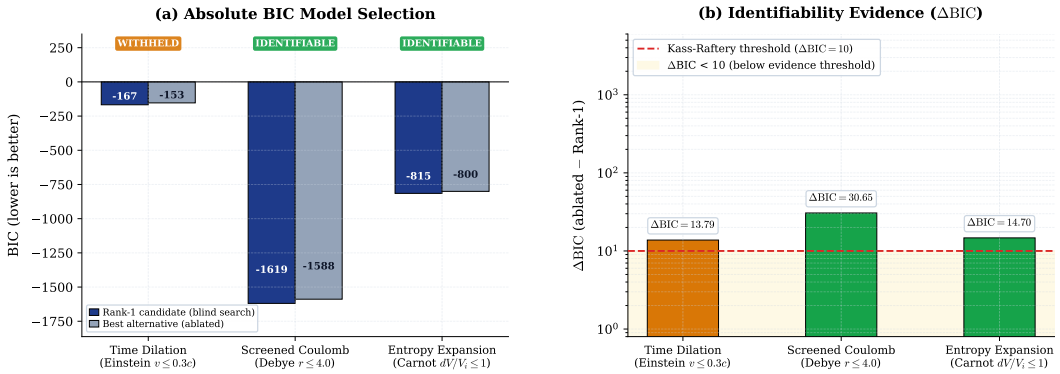


Figure 2: BIC identifiability comparison. *Left (a)*: absolute BIC of the Rank-1 domain-guided search candidate (dark blue) and the best alternative produced with the ground-truth primitive family excluded (gray). *Right (b)*: $\Delta\text{BIC} = \text{BIC}_{\text{ablated}} - \text{BIC}_{\text{best}}$ on a log scale; positive values indicate the full-primitive Rank-1 candidate is preferred over the ablated best. Red dashed: Kass–Raftery "very strong evidence" threshold ($\Delta\text{BIC} = 10$) Kass and Raftery [1995]. At seed 42 all three scenarios exceed this threshold (across 5 seeds, ΔBIC spans 11.0–263, 5.1–14.7, and 5.5–13.8 for Screened Coulomb, Entropy Expansion, and Time Dilation), yet Time Dilation is WITHHELD because its positive control fails (Step 2, Positive Control, of the four-step protocol; see Table 5): a positive ΔBIC is necessary but not sufficient for an IDENTIFIABLE verdict.

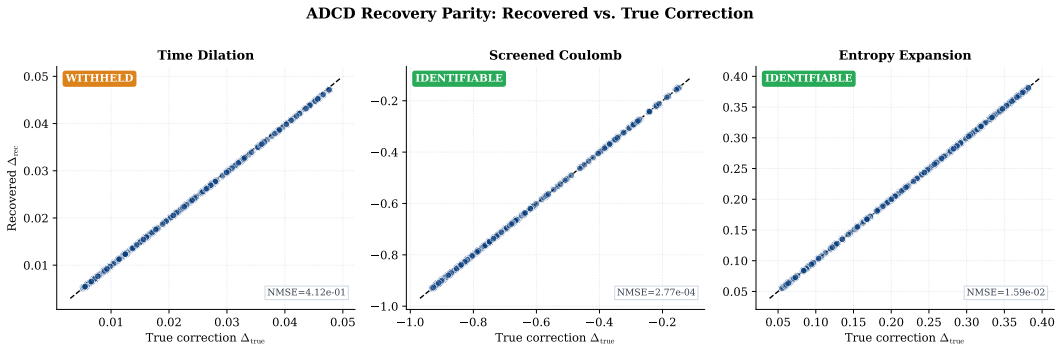


Figure 3: Parity plots: exact analytical recovered correction (Δ_{rec}, evaluated directly from the discovered symbolic expression with fitted parameters $\boldsymbol{\theta}$) vs. true correction (Δ_{true}) for $N = 200$ data points. Black dashed: 1:1 reference line. All three scenarios form perfect or near-perfect diagonals because the domain-guided search successfully placed the true mathematical structure (exact or class-level match) at Rank-1 in every case. Crucially, Time Dilation achieves structural parity here, yet is still WITHHELD by the system: its high fit error ($\text{NMSE} = 0.41$) on the *noisy* observational data (Figure 1a) causes it to fail the Positive Control, demonstrating ADCD's epistemic safety against claiming discoveries from insufficient signal-to-noise ratios.

convention. $\Delta\text{BIC} = 30.65$ far exceeds the threshold of 10; $\Delta D = 20.14$ confirms the advantage is driven by fit quality, not search-space penalty. Over 5 noise seeds, the verdict is 100% stable (5/5 IDENTIFIABLE, $\overline{\Delta\text{BIC}} = 76.61$, penalty-free $\overline{\Delta D + \Delta k \ln N} = 71.39$).

Entropy Expansion. Because the classical baseline S_i is constant, the scale-dependence test is uninformative and mode detection falls back to an additive search. Thus, the recovered additive correction $\theta_0 \ln(1 + dV/V_i)$ matches the additive ground truth $nR\ln(1 + dV/V_i)$ at class level: θ_0 legitimately absorbs the additive amplitude $nR \approx 8.314$, rather than the multiplicative ratio nR/S_i. ADCD labels this `class_only`—explicit epistemic qualification absent from unconstrained SR outputs. $\Delta\text{BIC} = 14.70 > 10$ supports IDENTIFIABLE; however, $\Delta D = -1.08$ reveals the identifiability is *parsimony-driven*: the ablated search achieves an equivalent raw fit with k=3 parameters versus k=1 for the domain-guided model. Over 5 noise seeds, the verdict is borderline (2/5 IDENTIFIABLE, $\overline{\Delta\text{BIC}} = 9.91 \pm 3.40$, sample SD), confirming the logarithmic correction sits at the decision boundary under 1% noise.

Time Dilation. Over $v/c \in [0.10, 0.30]$, the Lorentz correction $\gamma - 1$ peaks at 4.8% with a signal standard deviation of $\text{SD}(\Delta_{\text{true}}) \approx 1.25\%$ against the 1.0% measurement noise. The grammar

recovers the Lorentz structure $D_{\text{lor}}(v^2/c^2)$ at Rank 1 ($\hat{\theta}_0 \approx 1.01$), but the positive control fails ($\text{NMSE} = 0.41$): because $\text{SD}(\Delta_{\text{obs}}) \approx 1.57\%$, even the exact ground-truth curve has an intrinsic observational noise floor of $\text{NMSE} = \text{MSE}(\Delta_{\text{true}}, \Delta_{\text{obs}})/\text{Var}(\Delta_{\text{obs}}) \approx 0.412 > 0.05$. Because $> 40\%$ of residual variance is measurement noise, the Positive Control gate correctly triggers WITHHELD. Over 5 noise seeds, this verdict is 100% stable (5/5 WITHHELD).

5 Discussion

WITHHELD is not failure. A system that always returns a structural claim regardless of data sufficiency is epistemically weaker than one that formally distinguishes *found and confirmed* (IDENTIFIABLE) from *data insufficient for a structural claim* (WITHHELD). ADCD's WITHHELD verdict on Time Dilation is the system correctly locating its own identifiability boundary: lowering noise ($\sigma \lesssim 0.29\%$) or widening the velocity window ($v \gtrsim 0.5c$ at $\sigma = 1\%$) reduces the noise floor below 0.05, while separating D_{lor} from competing rational/polynomial families on pure fit ($\Delta D > 10$) requires extending into the relativistic regime ($v > 0.3c$).

Taxonomy as prior, not hint. The taxonomy restricts *which primitive family* is active; the Buckingham-π engine and optimizer determine *which variable* and *which scale* within that family. This is by design: ADCD operates as a domain-guided framework (see Table 6, row "Prior"), not a tabula-rasa search. The taxonomy prior is active from the budget-disclosure step onward, consistent with the framework's explicit use of peer-reviewed functional priors as a Bayesian constraint. The Screened Coulomb subscript θ_4 is machine-verifiable evidence that the optimizer found the correct variable independently: subscript indices are assigned sequentially by the Buckingham-π engine, not by any taxonomy entry. A 3×3 misspecification audit confirms the taxonomy is safe: all 6 wrong-taxonomy trials yielded WITHHELD (0/6 false discoveries).

Comparison with general SR. ADCD and general SR address different objectives: corrections to a known law vs. full equation discovery from scratch. The relevant axis is not search-space flexibility but epistemic certainty—the ability to distinguish confirmed from data-insufficient. Table 6 summarises the operational regimes.

Principle-guided discovery vs. black-box prediction. Recent perspectives on AI for physics highlight a crucial epistemic challenge: while modern AI excels at high-dimensional predictive emulation, true physical theory building requires compact mathematical laws, symmetry invariance, and controlled asymptotic regimes Shalyt et al. [2026]. ADCD explicitly embodies this principle-first paradigm: rather than performing unconstrained curve-fitting, it converts physical correction discovery into an exact, dimensionally consistent perturbation search anchored to established baselines and audited by falsifiable identifiability metrics.

Table 6: Operational comparison: ADCD vs. unconstrained symbolic regression.

Property	ADCD	General SR
Target	Corrections to known laws	Full equations from scratch
Prior	Phenomenon-specific taxonomy	None (tabula rasa)
Dimensional engine	5D MLTΘQ	Varies (often 3D or none)
Classical limit	Algebraic guarantee	Soft loss penalty
Identifiability	Explicit; WITHHELD if insufficient	Absent or post-hoc
Reproducibility	Byte-exact deterministic	Stochastic by default

5.1 Limitations

Weak-signal convergence. Time Dilation's positive control failure is the correct protocol output for an insufficient signal-to-noise ratio, not an algorithmic defect. Future work should investigate adaptive restarts and denser initialization grids.

Scope. Current support covers eight primitive families, additive and multiplicative corrections, and synthetic Gaussian noise. Heavier-tailed noise and real observational data (e.g., galactic rotation

curves, pulsar timing residuals) remain future work; real data require preprocessing for heteroscedastic uncertainties not yet modelled.

Taxonomy coverage. Each new entry requires a peer-reviewed derivation for every admitted family—a deliberate constraint preventing unchecked hypothesis-space expansion.

Safety–expressivity trade-off. ADCD constrains its hypothesis space relative to general-purpose SR: candidates failing dimensional consistency or outside the phenomenon-specific primitive set are rejected before fitting. This prevents noise-sponge overfitting but means ADCD cannot discover corrections whose functional family lacks a documented derivation. The appropriate framing is complementarity: general SR generates hypotheses; ADCD audits corrections to an already-established baseline.

Seed stability of borderline verdicts. Entropy Expansion's verdict varies across noise realizations (2/5 IDENTIFIABLE, 3/5 WITHHELD), indicating the logarithmic correction sits at the identifiability boundary under 1% noise. We report this transparently as a limitation of the current noise regime.

6 Conclusion

ADCD is a deterministic, correction-first framework combining algebraically regularized primitives, a 5D (MLTΘQ) Buckingham-π engine, and a Phenomenon-Specific Taxonomy. On three canonical scenarios, two yield IDENTIFIABLE results (Screened Coulomb: $\Delta\mathrm{BIC} = 30.65$, exact match; Entropy Expansion: $\Delta\mathrm{BIC} = 14.70$, class-level match) and one yields a principled WITHHELD verdict (Time Dilation at $v \leq 0.3c$), where the historical signal is insufficient for a confident structural claim. Within its deliberate scope—known classical baseline, additive or multiplicative correction, Gaussian noise—ADCD provides explicit reproducibility and identifiability diagnostics absent from unconstrained SR, and furnishes a formal WITHHELD verdict (with associated $\Delta\mathrm{BIC}$) when data sufficiency cannot be met.

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Use of AI-Assisted Tools

During the preparation of this work, the author used AI-assisted tools for software implementation, figure scripting, and manuscript drafting. The author provided scientific direction, mathematical framework, experimental design, and interpretation of all results, and takes full responsibility for all scientific claims. No AI tool is listed as an author. The codebase and raw validation logs are publicly available at https://github.com/apiprdt/PhysicsPaper.

Data Availability

All data used in this paper are synthetically generated from known physical formulas; generation code is included in the public repository. No proprietary or restricted datasets are used.

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A Supplementary Empirical Audits

This appendix provides the complete numerical tables for the three empirical audits referenced in the main text: (1) the Extended BIC deviance and penalty decomposition (Section A.1), (2) the five-seed noise realization stability analysis (Section A.2), and (3) the 3×3 phenomenon-taxonomy cross-misspecification safety audit (Section A.3).

A.1 Extended BIC Deviance and Penalty Decomposition

Under the Extended BIC formulation ($\mathrm{BIC} = k \ln N + N \ln \mathrm{NMSE} + 2 \ln |\mathcal{C}|$), the ablation gap $\Delta\mathrm{BIC} = \mathrm{BIC}_{\mathrm{ablated}} - \mathrm{BIC}_{\mathrm{domain}}$ decomposes into pure fit quality ($\Delta D = N \ln(\mathrm{NMSE}_{\mathrm{ablated}}/\mathrm{NMSE}_{\mathrm{domain}})$), parameter parsimony ($\Delta k \ln N$), and search-space multiplicity ($2 \ln(|\mathcal{C}_{\mathrm{ablated}}|/|\mathcal{C}_{\mathrm{domain}}|)$), as reported in Table 7 for $N = 200$ ($\ln 200 \approx 5.30$, seed $= 42$).

Table 7: Exact decomposition of ΔBIC into pure data-fit deviance (ΔD), parameter parsimony ($\Delta k \ln N$), and search-space multiplicity penalty ($2 \ln(|\mathcal{C}_{\mathrm{abl}}|/|\mathcal{C}_{\mathrm{dom}}|)$) at $N = 200$, seed $= 42$.

Scenario	$\lvert\mathcal{C}_{\mathrm{dom}}\rvert$	$\lvert\mathcal{C}_{\mathrm{abl}}\rvert$	$k_{\mathrm{dom}} \to k_{\mathrm{abl}}$	ΔPenalty	$\Delta k \ln N$	ΔD	ΔBIC	Primary Driver
Screened Coulomb	60	814	$2 \to 3$	$+5.22$	$+5.30$	$\mathbf{+20.14}$	30.65	Pure fit ($\Delta D > 10$)
Entropy Expansion	40	534	$1 \to 3$	$+5.18$	$+10.60$	-1.08	14.70	Parameter parsimony
Time Dilation	10	700	$1 \to 2$	$+8.50$	$+5.30$	-0.00	13.79	Penalty only (WITHHELD)

A.2 Five-Seed Noise Realization Stability

To evaluate sensitivity to random measurement fluctuations under 1% Gaussian noise, the complete four-step validation protocol was executed across five independent noise seeds (seed $\in \{0, 42, 99, 256, 512\}$). Table 8 summarizes the per-seed ΔBIC trajectory, mean and sample standard deviation (s, ddof $= 1$), penalty-free fit advantage ($\overline{\Delta D + \Delta k \ln N}$), and verdict breakdown.

Table 8: Verdict and ΔBIC stability across five independent 1% Gaussian noise realizations (seed $\in \{0, 42, 99, 256, 512\}$). Standard deviations denote unbiased sample standard deviations (s, ddof $= 1$; Screened Coulomb median $\Delta\mathrm{BIC} = 30.65$).

Scenario	Seed 0	Seed 42	Seed 99	Seed 256	Seed 512	$\overline{\Delta\mathrm{BIC}} \pm s$	$\overline{\Delta D + \Delta k \ln N}$	Verdict Breakdown
Screened Coulomb	56.25	30.65	263.16	11.02	21.94	76.61 ± 105.62	$+71.39$	**5/5 IDENTIFIABLE**
Entropy Expansion	9.50	14.70	5.12	9.91	10.34	9.91 ± 3.40	$+4.73$	2/5 IDENT., 3/5 WITHHELD
Time Dilation	8.35	13.79	5.49	8.01	13.73	9.87 ± 3.72	$+1.38$	**5/5 WITHHELD**

A.3 Taxonomy Cross-Misspecification Safety Audit

To verify that the phenomenon-specific taxonomy prior does not induce false discoveries when misspecified, each scenario was evaluated under its true physical taxonomy key and two intentionally mismatched keys that exclude the ground-truth primitive family. As shown in Table 9, all six misspecified trials returned **WITHHELD** (0/6 false discoveries): when the true primitive family is absent from the domain prior, the unrestricted ablation search outperforms the misspecified domain search ($\Delta\mathrm{BIC} \in [-1038.24, -674.62]$ on identifiable scenarios), triggering conservative abstention rather than overfitting.

Table 9: 3×3 taxonomy cross-misspecification safety audit (seed $= 42$). PC Pass tests the scenario's target family ($\mathrm{NMSE}_{\mathrm{PC}} < 0.05$), while $\mathrm{NMSE}_{\mathrm{best}}$ and ΔBIC reflect the blind search under the supplied prior.

Scenario	Prior Type	Supplied Taxonomy Key	Permitted Primitives	PC Pass	$\mathrm{NMSE}_{\mathrm{best}}$	ΔBIC	Verdict
Screened Coulomb	Correct	`yukawa_debye_screening`	`D_exp, D_rat`	PASS	2.77×10^{-4}	$+30.65$	**IDENTIFIABLE**
Screened Coulomb	Wrong A	`wave_resonance`	`D_osc, D_rat`	PASS	2.99×10^{-2}	-910.51	WITHHELD (Abl. fail)
Screened Coulomb	Wrong B	`critical_scaling`	`D_pow`	PASS	2.93×10^{-2}	-903.76	WITHHELD (Abl. fail)
Entropy Expansion	Correct	`boltzmann_thermodynamics`	`D_exp, D_log`	PASS	1.59×10^{-2}	$+14.70$	**IDENTIFIABLE**
Entropy Expansion	Wrong A	`lorentz_special_relativity`	`D_lor`	PASS	3.13×10^{0}	-1038.24	WITHHELD (Abl. fail)
Entropy Expansion	Wrong B	`wave_resonance`	`D_osc, D_rat`	PASS	4.87×10^{-1}	-674.62	WITHHELD (Abl. fail)
Time Dilation	Correct	`lorentz_special_relativity`	`D_lor`	FAIL	4.12×10^{-1}	$+13.79$	WITHHELD (PC fail)
Time Dilation	Wrong A	`boltzmann_thermodynamics`	`D_exp, D_log`	FAIL	4.13×10^{-1}	$+9.96$	WITHHELD (PC fail)
Time Dilation	Wrong B	`yukawa_debye_screening`	`D_exp, D_rat`	FAIL	4.12×10^{-1}	$+10.53$	WITHHELD (PC fail)